# Mastering the Azure Provider in Terraform

Learning how to use Terraform with Azure saves you from tedious manual clicking in the web portal. It lets you build and destroy entire infrastructures using simple, reusable code files.

**What you'll learn:**

1. Connecting to Azure with the Provider
2. Preparing Your Folder with Init
3. Previewing with Terraform Plan
4. Deploying with Terraform Apply

**Aman Raj Singh**

Cloud Operations Engineer @ iManage

linkedin.com/in/mramanrajsingh

1

TIP 1 OF 4

## Connecting to Azure with the Provider

Before Terraform can build anything for you, it needs to know which cloud you are using. You define the Azure provider in your configuration file so Terraform downloads the right plugin to communicate with Microsoft Azure.

```
terraform {
  required_providers {
    azurerm = {
      source  = "hashicorp/azurerm"
      version = "~> 3.0"
    }
  }
}

provider "azurerm" {
  features {}
}
```

Swipe to tip 2 >> linkedin.com/in/mramanrajsingh

2

TIP 2 OF 4

## Preparing Your Folder with Init

Every time you start a new Terraform project, your folder is empty. Running the init command scans your code files, figures out that you are using Azure, and downloads the required provider plugins into a hidden folder.

```
$terraform init
```

Swipe to tip 3 >> linkedin.com/in/mramanrajsingh

3

TIP 3 OF 4

## Previewing with Terraform Plan

Never run blind in the cloud. The plan command compares your code with what currently exists in your Azure account. It gives you a clear list of what will be added, changed, or deleted so you can catch mistakes early.

```
$terraform plan
```

Swipe to tip 4 >> linkedin.com/in/mramanrajsingh

4

TIP 4 OF 4

## Deploying with Terraform Apply

This is where the magic happens. When you run apply, Terraform takes your code and starts building the real resources in your Azure subscription. Type 'yes' when prompted, and your cloud infrastructure is ready in minutes.

```
$terraform apply
```

See Key Takeaways on next page >> linkedin.com/in/mramanrajsingh

# Key Takeaways

+ Always run terraform plan before applying changes to production
+ Keep your state files secure and never share them publicly
+ Use descriptive names for your resource groups and virtual networks
+ Break your infrastructure code into smaller, logical files
+ Clean up unused resources with terraform destroy to save money

## Watch Out For

! Hardcoding passwords or secret keys directly into your configuration files
! Forgetting to run terraform init before trying to run a plan or apply
! Modifying Azure resources manually in the portal while also managing them with Terraform

**Follow for daily Cloud & DevOps guides!**

linkedin.com/in/mramanrajsingh